

Privacy-Safe Voter Inclusion Proofs

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Abstract

Voters may reasonably want assurance that an accepted or counted record exists without receiving a receipt that proves how they voted. This paper defines the baseline RCOUNT privacy gate for inclusion-style proofs. A public proof may carry an anonymized token hash and a reporting-unit reference, but it must not carry candidate selections, direct voter identity, ballot style, timestamps, or other linkable facts that create coercion or vote-buying risk. Using the `privacy-inclusion-sketch` and `choice-bearing-proof` fixtures, we show how RCOUNT can reject choice-bearing artifacts while preserving a narrow path toward receipt-safe public verification.

1 Introduction

Public verification has a sharp privacy edge. A voter should be able to ask whether a counted or accepted record is present without gaining an artifact that can prove a candidate choice to a buyer, employer, family member, campaign, or state actor. The hard rule for RCOUNT is therefore simple: prove inclusion, never prove choices.

V.3 described how source and package hashes make public evidence tamper-evident. V.4 narrows the problem to voter-facing proof material. The baseline proof is not a complete cryptographic voting system. It is a privacy gate over public proof records: if the record carries candidate selections or linkable voter facts, it fails.

The voter-facing interpretation is deliberately narrow. A passing proof can support the statement that an anonymized accepted-token or counted-token record appears in the public package. It does not support the stronger statement that “my vote for candidate X was counted,” because that stronger statement is exactly the receipt a coercer would want. Refusing to publish that evidence is a voter protection, not a missing feature.

2 Proof Model

The current RCOUNT proof sketch has a deliberately small public surface:

Field	Public meaning
<code>proof_id</code>	Stable package-local proof identifier.
<code>kind</code>	Accepted-token or counted-token proof kind.
<code>token_hash</code>	Anonymized token digest, not a candidate-choice digest.
<code>reporting_unit_id</code>	Public reporting unit associated with the proof.
<code>candidate_selections</code>	Must be empty in the baseline public proof.
<code>voter_id, ballot_style, issued_at</code>	Must be absent in public baseline proofs.

This is a public proof contract, not a complete token-issuance protocol:

Contract point	Requirement
Identifier scope	<code>proof_id</code> is package-local and should not be a durable voter identifier.
Allowed evidence	The public record may expose proof kind, anonymized token hash, and coarse reporting-unit reference.
Forbidden evidence	Candidate selections, voter identity, ballot style, precise timestamps, device identifiers, and equivalent linkers are outside the baseline public proof.
Token caveat	A hash is not privacy by itself; token issuance, salts, nonces, key custody, and disclosure policy remain protocol work.
Gate meaning	Passing <code>proof_privacy_gate</code> is a necessary baseline check, not a formal coercion-resistance proof.

The distinction between accepted and counted matters. An accepted-token proof follows ballot acceptance decisions: the voter returned material, the office resolved eligibility and cure questions under local law, and the package represents an anonymized accepted-record token. A counted-token proof follows counting and any adjudication path represented in the package. Neither form says which candidates were selected, whether a recount will later change the public record, or whether a canvassing board has certified final totals.

The token is also not the same thing as a voter, ballot card, ballot, CVR row, contest row, or vote. Those accounting relationships belong to the larger RCOUNT evidence model. V.5 handles ballot manifests and V.6 handles CVR-to-summary reconciliation; V.4 only defines the receipt-safe public edge of an inclusion-style proof.

3 Privacy Gate

The `proof_privacy_gate` is intentionally conservative. It rejects a proof if it exposes candidate selections, combines voter identity with ballot style and timestamps, or uses an invalid token hash. These are mechanical gates, not a full privacy proof.

Actor demand	Why the baseline proof should not satisfy it
Vote buyer asks for proof of a selected candidate	Candidate selections are absent and any choice-bearing proof fails.
Employer asks a worker to reveal a public proof id	The identifier is package-local and should not be a durable identity token.
Campaign combines ballot style and issue time	Ballot style and precise timestamps are forbidden in the public proof.
Observer targets a tiny reporting unit or rare contest	Small-cell and rare-contest metadata require disclosure review before publication.
Insider tries deterministic token linkage across exports	Token issuance, salts, nonces, and custody are protocol controls outside this public gate.

Rejected public field pattern	Risk
Candidate selections in proof	Creates a direct vote-choice receipt.
Voter identity in proof	Links public proof material to a person.
Ballot style plus timestamp	May reidentify a voter in small populations.
Small reporting unit plus rare contest facts	Can create small-cell disclosure risk.

Passing the gate means only that the baseline public record avoids these obvious receipt channels. It does not prove end-to-end coercion resistance.

The proof surface can be visualized as a narrow public window:



choices, voter identity, ballot style, precise time, and device facts stay outside

4 Fixtures

The positive fixture is **privacy-inclusion-sketch**. It contains one accepted-token proof with an anonymized token hash and no candidate selections:

```

{
  "proof_id": "proof:accepted-token-001",
  "kind": "anonymized-accepted-ballot-token",
  "token_hash": "sha256:aaaaaaaa...",
  "reporting_unit_id": "syn:precinct:P-001",
  "candidate_selections": []
}
  
```

The negative fixture is **choice-bearing-proof**. It changes the same proof by adding **cand-a** to **candidate_selections**. That proof must fail, even if package arithmetic and source hashes pass.

Fixture	Proof file	Transcript file
privacy-inclusion-sketch	proof NDJSON	verify transcript
choice-bearing-proof	proof NDJSON	verify transcript

In both packages those anchors are the package-local paths `proofs/inclusion-proofs.ndjson` and `transcripts/verify-transcript.json`.

The pair catches a concrete regression: if a future verifier accepts a public proof that contains candidate selections, the negative fixture should fail the acceptance run.

5 Verifier Traceability

The fixtures can be regenerated and checked directly:

```
cargo run -p rcount-io --example privacy_inclusion_sketch_package
cargo run -p rcount-io --example choice_bearing_proof_package
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/privacy-inclusion-sketch
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/choice-bearing-proof
```

Fixture	Expected result	Carrying check
privacy-inclusion-sketch	transcript status pass	<code>proof_privacy_gate</code> passes for <code>proof:accepted-token-001</code> ; <code>source_hash_match</code> also passes.
choice-bearing-proof	transcript status fail	<code>proof_privacy_gate</code> fails with <code>proof proof:accepted-token-001</code> exposes candidate selections; <code>source_hash_match</code> still passes.

The negative transcript anchors the failure to the privacy gate:

```
{
  "equation_id": "proof_privacy_gate",
  "status": "fail",
  "error": "proof proof:accepted-token-001 exposes candidate selections"
}
```

That separation matters. The package can be well-formed, hash-bound, and arithmetically coherent while still being unacceptable for public voter-facing proof use.

6 Boundaries

RCOUNT does not decide voter eligibility. Eligibility, cure, provisional adjudication, and ballot acceptance are legal and administrative decisions. A proof can refer to an accepted or counted token only after those decisions have been represented in the package.

The ordinary lifecycle is returned, accepted or rejected, counted if eligible and countable, canvassed, and certified. Jurisdictions vary in cure windows, provisional-ballot handling, recount procedures, public-record rules, and voter-verification rights. V.4 therefore defines a minimum receipt-safety rule for public artifacts, not a national canvass workflow.

RCOUNT also does not give voters a candidate-choice receipt. A public artifact that proves a voter selected a candidate is a failure, not a feature. The baseline proof sketch avoids candidate selections by construction.

Reporting-unit references need care. Vote centers, central-count mail batches, split precincts, and batch boundaries may be less intuitive than a voter-facing precinct label. In small units, even coarse metadata can become sensitive when combined with rare contests, write-ins, ballot styles, or timing facts.

Finally, V.4 is not a complete end-to-end voting protocol. It does not specify ballot casting, voter authentication, mixnets, cryptographic tallying, or a formal coercion-resistance proof. It also does not specify token salts, nonces, key custody, detached signatures, or issuance ceremonies. Those would require a separate protocol paper. V.4 only defines the minimum privacy gate RCOUNT public proof records must pass.

7 Conclusion

Privacy-safe inclusion is a narrow but valuable promise. RCOUNT can let a public package carry anonymized accepted or counted token evidence, but the package must not carry candidate choices, direct voter identity, or linkable small-cell facts. The positive and negative fixtures establish the baseline: an anonymized token can pass, while a choice-bearing proof must fail.

That line protects the rest of the V track. Public verification should make counts more accountable without making voters more coercible. The next papers can add ballot manifest and CVR reconciliation evidence on the accounting side, but V.4 fixes the public privacy invariant they must preserve.

References

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